

Relatório do trabalho de grupo — Tópico 5

Plano de Monitorização, Backup e Continuidade Operacional

Módulo: Linux, Segurança e Cloud

1. Identificação do grupo

Campo	Conteúdo
Grupo	Grupo X
Elementos	Kleiton Renato - João Sousa - Eliézer Santos - Samori Gomes - Érica Lopes
Papeis assumidos	Análise da superfície de ataque; Regras de firewall; Hardening e validação; Documentação

2. Serviço Analisado

Serviço escolhido

Foi seleccionado o cenário **HTML com Nginx**, utilizando uma máquina virtual Ubuntu Linux com o servidor web Nginx para disponibilizar uma página HTML estática.

Componentes críticos

- Sistema Operativo Ubuntu
- Servidor Web Nginx
- Ficheiros HTML do site
- Configuração do Nginx
- Sistema de ficheiros do servidor

Ficheiros e configurações importantes

Componente	Localização
Página principal	/var/www/html/index.html
Configuração principal do Nginx	/etc/nginx/nginx.conf
Sites configurados	/etc/nginx/sites-available/
Sites ativos	/etc/nginx/sites-enabled/

Logs relevantes

Log	Localização
Access Log	/var/log/nginx/access.log
Error Log	/var/log/nginx/error.log
Logs do sistema	journalctl

Riscos de indisponibilidade

- Falha do serviço Nginx
 - Remoção acidental dos ficheiros HTML
 - Falta de espaço em disco
 - Erros de configuração
 - Atualizações mal sucedidas
 - Acesso não autorizado ao servidor
-

3. Plano de Monitorização

Elemento a monitorizar	Como Verificar	Frequências sugerida	Evidência
Estado do serviço web	systemctl status nginx	Diariamente	Captura de ecrã
Espaço em disco	df -h	Diariamente	Captura de ecrã
Memória	free -h	Diariamente	Captura de ecrã
Logs de erro	tail -f /var/log/nginx/error.log	Diariamente	Registo dos eventos
Autenticação	cat /var/log/auth.log	Semanalmente	Captura de ecrã

Comandos utilizados

Para ver estado do serviço web => systemctl status nginx

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-06-10 06:36:16 -01; 13h ago
 Invocation: 80a9b5ee33ff45d595180dfc651f1307
    Docs: man:nginx(8)
   Process: 22259 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 22261 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Main PID: 22286 (nginx)
    Tasks: 2 (limit: 298)
   Memory: 2.3M (peak: 5.3M)
      CPU: 72ms
   CGroup: /system.slice/nginx.service
           └─22286 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─22289 "nginx: worker process"

jun 10 06:36:16 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server.
jun 10 06:36:16 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
lines 1-17/17 (END)
```

Para ver espaço em disco => df -h

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$ df -h
Sistema de ficheiros Tamanho  Uso Livre  Uso% Montado em
tmpfs                316M  2,2M   314M    1% /run
/dev/sda2             27G   13G   13G    51% /
tmpfs                789M   8,0K   789M    1% /dev/shm
efivarfs             128M   32K   128M    1% /sys/firmware/efi/efivars
none                 1,0M    0    1,0M    0% /run/credentials/systemd-journald.service
tmpfs                789M   57M   733M    8% /tmp
none                 1,0M    0    1,0M    0% /run/credentials/systemd-resolved.service
/dev/sda1             1,1G   6,4M   1,1G    1% /boot/efi
tmpfs                158M   88K   158M    1% /run/user/1000
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$
```

Memória => free -h

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3,7Gi       878Mi       1,8Gi        48Mi       1,2Gi       2,9Gi
Swap:          4,0Gi       138Mi       3,9Gi
```

Para ver Logs de erro => tail -20 /var/log/nginx/error.log

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$ tail -f /var/log/nginx/error.log
2026/06/10 06:36:17 [notice] 22286#22286: using inherited sockets from "5;"
```

Para ver Autenticação => cat /var/log/auth.log

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$ cat /var/log/auth.log
2026-06-08T17:04:53.142924-01:00 kleiton-da-rosa-Virtual-Machine gdm-password: pam_unix(gdm-password:auth): conversation failed
2026-06-08T17:05:40.504951-01:00 kleiton-da-rosa-Virtual-Machine pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:05:40.514035-01:00 kleiton-da-rosa-Virtual-Machine pkexec[14817]: kleiton-da-rosa: Executing command [USER=root] [TTY=unknown] [CWD=/home/kleiton-da-rosa] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-08T17:07:19.083036-01:00 kleiton-da-rosa-Virtual-Machine sshd-session[15237]: Accepted password for kleiton-da-rosa from 192.168.1.164 port 64334 ssh2
2026-06-08T17:07:19.098872-01:00 kleiton-da-rosa-Virtual-Machine sshd-session[15237]: pam_unix(sshd:session): session opened for user kleiton-da-rosa(uid=1000) by kleiton-da-rosa(uid=0)
2026-06-08T17:07:19.106098-01:00 kleiton-da-rosa-Virtual-Machine systemd-logind[1182]: New session '78' of user 'kleiton-da-rosa' with class 'user' and type 'tty'.
2026-06-08T17:08:27.630917-01:00 kleiton-da-rosa-Virtual-Machine pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:08:27.635852-01:00 kleiton-da-rosa-Virtual-Machine pkexec[15846]: kleiton-da-rosa: Executing command [USER=root] [TTY=unknown] [CWD=/home/kleiton-da-rosa] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-08T17:09:01.932367-01:00 kleiton-da-rosa-Virtual-Machine CRON[15921]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-08T17:09:01.945410-01:00 kleiton-da-rosa-Virtual-Machine CRON[15921]: pam_unix(cron:session): session closed for user root
2026-06-08T17:11:26.086869-01:00 kleiton-da-rosa-Virtual-Machine pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:11:26.089861-01:00 kleiton-da-rosa-Virtual-Machine pkexec[15941]: kleiton-da-rosa: Executing command [USER=root] [TTY=unknown] [CWD=/home/kleiton-da-rosa] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-08T17:17:01.966057-01:00 kleiton-da-rosa-Virtual-Machine CRON[15956]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
```

4. Plano de Logs

Log	Finalidade	Evento relevante
journalctl	Monitorizar serviços do sistema	Falha no Nginx
auth.log	Verificar acessos e autenticação	Tentativas de login inválidas
access.log	Registrar acessos ao site	Número de visitas
error.log	Identificar erros da aplicação	Erros HTTP 404 e 500

Exemplos de consulta

journalctl

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária/skodjidental/Linux-e-Seguran-a-na-Cloud/topico-05
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjidental/Linux-e-Seguran-a-na-Cloud/topico-05$ journalctl
mai 27 18:02:13 localhost.localdomain kernel: Linux version 7.0.0-14-generic (builddd@lcy02-amd64-043) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1)
mai 27 18:02:13 localhost.localdomain kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-14-generic root=UUID=58ffbf9-2788-4d7a-81e0-14a29b0cc35a
mai 27 18:02:13 localhost.localdomain kernel: KERNEL supported cpus:
mai 27 18:02:13 localhost.localdomain kernel: Intel GenuineIntel
mai 27 18:02:13 localhost.localdomain kernel: AMD AuthenticAMD
mai 27 18:02:13 localhost.localdomain kernel: Hygon HygonGenuine
mai 27 18:02:13 localhost.localdomain kernel: Centaur CentaurHauls
mai 27 18:02:13 localhost.localdomain kernel: zhaoxin Shanghai
mai 27 18:02:13 localhost.localdomain kernel: BIOS-provided physical RAM map:
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000009fff] System RAM
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [gap 0x000000000000a0000-0x000000000000bffff]
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x000000000000c0000-0x000000000000ffff] device reserved
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000000100000-0x00000000007ff40fff] System RAM
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000007ff41000-0x00000000007ff8fff] device reserved
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000007ff9000-0x00000000007ffaffff] ACPI data
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000007ffb000-0x00000000007ffeffff] ACPI NVS
mai 27 18:02:13 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000007ffff000-0x00000000007fffffff] System RAM
mai 27 18:02:13 localhost.localdomain kernel: NX (Execute Disable) protection: active
mai 27 18:02:13 localhost.localdomain kernel: APIC: Static calls initialized
mai 27 18:02:13 localhost.localdomain kernel: efi: EFI v2.7 by Microsoft
mai 27 18:02:13 localhost.localdomain kernel: efi: ACPI=0x7fffa000 ACPI 2.0=0x7fffa014 SMBIOS=0x7ff85000 SMBIOS 3.0=0x7ff83000 MEMATTR=0x7ef06a98 MO
mai 27 18:02:13 localhost.localdomain kernel: random: crng init done
mai 27 18:02:13 localhost.localdomain kernel: SMBIOS 3.1.0 present.
mai 27 18:02:13 localhost.localdomain kernel: DMI: Microsoft Corporation Virtual Machine/Virtual Machine, BIOS Hyper-V UEFI Release v4.1 06/20/2024
mai 27 18:02:13 localhost.localdomain kernel: DMI: Memory slots populated: 1/1
mai 27 18:02:13 localhost.localdomain kernel: secureboot: Secure boot disabled
mai 27 18:02:13 localhost.localdomain kernel: Hypervisor detected: Microsoft Hyper-V
mai 27 18:02:13 localhost.localdomain kernel: Hyper-V: privilege flags low 0x2e7f, high 0x3b8030, ext 0x2, hints 0x42c2c, misc 0xbcd7b2
mai 27 18:02:13 localhost.localdomain kernel: Hyper-V: LAPIC Timer Frequency: 0x30d40
mai 27 18:02:13 localhost.localdomain kernel: Hyper-V: Using hypercall for remote TLB flush
mai 27 18:02:13 localhost.localdomain kernel: clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns
mai 27 18:02:13 localhost.localdomain kernel: clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440
mai 27 18:02:13 localhost.localdomain kernel: tsc: Marking TSC unstable due to running on Hyper-V
mai 27 18:02:13 localhost.localdomain kernel: tsc: Detected 1796.627 MHz processor
```

journalctl -u nginx

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária/skodjigital/Linux-e-Seguran-a-na-Cloud/topico-05
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjigital/Linux-e-Seguran-a-na-Cloud/topico-05$ journalctl -u nginx
jun 04 19:00:20 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 19:00:20 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server...
jun 04 19:10:21 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
jun 04 19:10:21 kleiton-da-rosa-Virtual-Machine systemd[1]: nginx.service: Deactivated successfully.
jun 04 19:10:21 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
jun 04 19:10:21 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 19:10:21 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
-- Boot ba9442e814e9483e8233d15c67445a72 --
jun 04 21:00:28 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:00:29 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine systemd[1]: nginx.service: Deactivated successfully.
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine nginx[5082]: 2026/06/04 21:49:46 [warn] 5082#5082: conflicting server name "-" on 0.0.0.0:80, ignored
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine nginx[5084]: 2026/06/04 21:49:46 [warn] 5084#5084: conflicting server name "-" on 0.0.0.0:80, ignored
jun 04 21:49:46 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
jun 04 21:55:24 kleiton-da-rosa-Virtual-Machine systemd[1]: Reloading nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:55:24 kleiton-da-rosa-Virtual-Machine nginx[5219]: 2026/06/04 21:55:24 [notice] 5219#5219: signal process started
jun 04 21:55:24 kleiton-da-rosa-Virtual-Machine systemd[1]: Reloaded nginx.service - A high performance web server and a reverse proxy server.
jun 04 21:56:52 kleiton-da-rosa-Virtual-Machine systemd[1]: Reloading nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:56:52 kleiton-da-rosa-Virtual-Machine nginx[5274]: 2026/06/04 21:56:52 [notice] 5274#5274: signal process started
jun 04 21:56:52 kleiton-da-rosa-Virtual-Machine systemd[1]: Reloaded nginx.service - A high performance web server and a reverse proxy server.
jun 04 21:57:02 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:57:02 kleiton-da-rosa-Virtual-Machine systemd[1]: nginx.service: Deactivated successfully.
jun 04 21:57:02 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
jun 04 21:57:29 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 21:57:29 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server...
jun 04 22:16:07 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
jun 04 22:16:07 kleiton-da-rosa-Virtual-Machine systemd[1]: nginx.service: Deactivated successfully.
jun 04 22:16:07 kleiton-da-rosa-Virtual-Machine systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
jun 04 22:16:07 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 22:16:07 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
jun 04 22:39:48 kleiton-da-rosa-Virtual-Machine systemd[1]: Reloading nginx.service - A high performance web server and a reverse proxy server...
lines 1-34, ...skipping...
jun 04 19:00:20 kleiton-da-rosa-Virtual-Machine systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
jun 04 19:00:20 kleiton-da-rosa-Virtual-Machine systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server...
```

cat /var/log/auth.log

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária/skodjigital/Linux-e-Seguran-a-na-Cloud/topico-05
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária/skodjigital/Linux-e-Seguran-a-na-Cloud/topico-05$ cat /var/log/auth.log
2026-06-08T17:04:53.142924-01:00 kleiton-da-rosa-Virtual-Machine gdm-password[auth]: conversation failed
2026-06-08T17:05:40.504951-01:00 kleiton-da-rosa-Virtual-Machine pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:05:40.514035-01:00 kleiton-da-rosa-Virtual-Machine pkexec[14817]: kleiton-da-rosa: Executing command [USER=root] [TTY=unknown] [CWD=/home/kleiton-da-rosa] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-08T17:07:19.083036-01:00 kleiton-da-rosa-Virtual-Machine sshd-session[15237]: Accepted password for kleiton-da-rosa from 192.168.1.164 port 64334 ssh
2026-06-08T17:07:19.088872-01:00 kleiton-da-rosa-Virtual-Machine sshd-session[15237]: pam_unix(sshd:session): session opened for user kleiton-da-rosa(uid=1000) by kleiton-da-rosa(uid=0)
2026-06-08T17:07:19.106098-01:00 kleiton-da-rosa-Virtual-Machine systemd-logind[1182]: New session '78' of user 'kleiton-da-rosa' with class 'user' and type 'tty'.
2026-06-08T17:08:27.630917-01:00 kleiton-da-rosa-Virtual-Machine pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:08:27.635852-01:00 kleiton-da-rosa-Virtual-Machine pkexec[15846]: kleiton-da-rosa: Executing command [USER=root] [TTY=unknown] [CWD=/home/kleiton-da-rosa] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-08T17:09:01.922367-01:00 kleiton-da-rosa-Virtual-Machine CRON[15921]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-08T17:09:01.945410-01:00 kleiton-da-rosa-Virtual-Machine CRON[15921]: pam_unix(cron:session): session closed for user root
2026-06-08T17:11:26.086869-01:00 kleiton-da-rosa-Virtual-Machine pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:11:26.089861-01:00 kleiton-da-rosa-Virtual-Machine pkexec[15941]: kleiton-da-rosa: Executing command [USER=root] [TTY=unknown] [CWD=/home/kleiton-da-rosa] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-08T17:17:01.966057-01:00 kleiton-da-rosa-Virtual-Machine CRON[15956]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-08T17:17:01.973423-01:00 kleiton-da-rosa-Virtual-Machine CRON[15956]: pam_unix(cron:session): session closed for user root
2026-06-08T17:30:02.002472-01:00 kleiton-da-rosa-Virtual-Machine CRON[15975]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-08T17:30:02.010732-01:00 kleiton-da-rosa-Virtual-Machine CRON[15975]: pam_unix(cron:session): session closed for user root
2026-06-08T17:39:01.043594-01:00 kleiton-da-rosa-Virtual-Machine CRON[15987]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-08T17:39:01.072224-01:00 kleiton-da-rosa-Virtual-Machine CRON[15987]: pam_unix(cron:session): session closed for user root
2026-06-08T17:51:10.943913-01:00 kleiton-da-rosa-Virtual-Machine sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:51:10.967817-01:00 kleiton-da-rosa-Virtual-Machine sudo: kleiton-da-rosa : TTY=/dev/pts/0 ; FWD=/home/kleiton-da-rosa ; USER=root ; COMMAND=/usr/bin/systemctl status nginx
2026-06-08T17:51:10.975401-01:00 kleiton-da-rosa-Virtual-Machine sudo: pam_unix(sudo:session): session closed for user root
2026-06-08T17:51:20.266817-01:00 kleiton-da-rosa-Virtual-Machine sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kleiton-da-rosa(uid=1000)
2026-06-08T17:51:20.267935-01:00 kleiton-da-rosa-Virtual-Machine sudo: kleiton-da-rosa : TTY=/dev/pts/0 ; FWD=/home/kleiton-da-rosa ; USER=root ; COMMAND=/usr/bin/systemctl status nginx
2026-06-08T17:53:48.697260-01:00 kleiton-da-rosa-Virtual-Machine sudo: pam_unix(sudo:session): session closed for user root
```

tail -20 /var/log/nginx/access.log

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: /var/www/wordpress$ tail -20 /var/log/nginx/access.log
192.168.1.67 - - [10/Jun/2026:20:31:59 -0100] "GET / HTTP/1.1" 302 5 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:00 -0100] "GET /wp-admin/setup-config.php HTTP/1.1" 200 4498 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:00 -0100] "GET /wp-includes/css/buttons.min.css?ver=7.0 HTTP/1.1" 304 0 "http://192.168.1.135/wp-admin/setup-config.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:00 -0100] "GET /wp-admin/css/forms.min.css?ver=7.0 HTTP/1.1" 304 0 "http://192.168.1.135/wp-admin/setup-config.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:00 -0100] "GET /wp-admin/css/install.min.css?ver=7.0 HTTP/1.1" 304 0 "http://192.168.1.135/wp-admin/setup-config.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:00 -0100] "GET /favicon.ico HTTP/1.1" 302 5 "http://192.168.1.135/wp-admin/setup-config.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:01 -0100] "GET /wp-admin/setup-config.php HTTP/1.1" 200 4498 "http://192.168.1.135/wp-admin/setup-config.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:17 -0100] "POST /wp-admin/setup-config.php?step=0 HTTP/1.1" 200 1213 "http://192.168.1.135/wp-admin/setup-config.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:17 -0100] "GET /favicon.ico HTTP/1.1" 302 5 "http://192.168.1.135/wp-admin/setup-config.php?step=0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:18 -0100] "GET /wp-admin/setup-config.php HTTP/1.1" 200 4498 "http://192.168.1.135/wp-admin/setup-config.php?step=0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:19 -0100] "GET /wp-admin/setup-config.php?step=1&language=pt_PT HTTP/1.1" 200 1943 "http://192.168.1.135/wp-admin/setup-config.php?step=0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:19 -0100] "GET /favicon.ico HTTP/1.1" 302 5 "http://192.168.1.135/wp-admin/setup-config.php?step=1&language=pt_PT" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
192.168.1.67 - - [10/Jun/2026:20:32:20 -0100] "GET /wp-admin/setup-config.php HTTP/1.1" 200 4498 "http://192.168.1.135/wp-admin/setup-config.php?step=1&language=pt_PT" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: /var/www/wordpress$
```

tail -20 /var/log/nginx/error.log

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$ tail -20 /var/log/nginx/error.log
2026/06/10 06:36:17 [notice] 22286#22286: using inherited sockets from "5";
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária/skodjdidigital/Linux-e-Seguran-a-na-Cloud/topico-05$
```

5. Plano de Backup

Itens a proteger

Item	Backup necessário?	Frequência	Justificação
Ficheiros do site	Sim	Diário	Recuperação rápida da página
Configuração web	Sim	Semanal	Restaurar configurações do Nginx
Base de dados	Não aplicável	Não aplicável	Site HTML não utiliza BD
Documentação	Sim	Semanal	Preservar procedimentos

Estratégia de Backup

- Frequência:
 - Site: diária
 - Configurações: semanal
 - Documentação: semanal
- Local de armazenamento:
 - Diretório de backup local
 - Repositório GitHub
- Retenção mínima:
 - Últimos 7 backups
- Responsável pela validação:
 - Administrador do sistema

Exemplo de backup

```
mkdir -p /backup
```

```
cp -r /var/www/html /backup/
```

```
cp -r /etc/nginx /backup/
```

Recuperação

```
cp -r /backup/html/* /var/www/html/
```

```
cp -r /backup/nginx/* /etc/nginx/
```

```
systemctl restart nginx
```

6. Plano de Recuperação

Cenário 1 – Página desapareceu

1. Verificar a existência do ficheiro index.html.
2. Consultar access.log e error.log.
3. Restaurar a partir do backup.
4. Reiniciar o Nginx.
5. Validar acesso através do navegador.

Cenário 2 – Serviço Nginx parado

1. Verificar o estado do serviço.

```
systemctl status nginx
```

2. Reiniciar o serviço.

```
systemctl restart nginx
```

3. Confirmar funcionamento.

```
systemctl status nginx
```

Cenário 3 – Disco cheio

1. Verificar utilização.

```
df -h
```

2. Remover ficheiros desnecessários.
3. Limpar logs antigos.
4. Confirmar libertação de espaço.

Validação da recuperação

- A página abre corretamente.
- O serviço Nginx está ativo.
- Não existem erros em error.log.
- O navegador devolve o HTTP 200.

Evidências a guardar

- Capturas de ecrã.
- Logs consultados.
- Registo da data e hora da recuperação.
- Relatório do incidente.

7. Riscos e Medidas de Mitigação

Risco	Mitigação
Apagamento de ficheiros	Backups diários
Falha do Nginx	Monitorização diária
Falta de espaço em disco	Verificação periódica
Erro humano	Documentação e procedimentos
Acesso indevido	Contas com sudo e palavras-passe fortes

8. Link do Repositório GitHub

Inserir o URL do repositório GitHub do grupo.

<https://github.com/Kleiton559/Linux-e-Seguran-a-na-Cloud>

Exemplo:

Criamos um exemplo de automatizar o arranque de um servidor nginx

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine: ~/Secretária
GNU nano 8.7.1 check_nginx.sh
#!/bin/bash

# Verifica o estado do Nginx
if systemctl is-active --quiet nginx; then
    echo "$(date) - Nginx está a funcionar."
else
    echo "$(date) - Nginx está parado. A iniciar..."

    # Tenta iniciar o Nginx
    systemctl start nginx

    # Verifica novamente
    if systemctl is-active --quiet nginx; then
        echo "$(date) - Nginx iniciado com sucesso."
    else
        echo "$(date) - Falha ao iniciar o Nginx."
    fi
fi

[ 18 linhas lidas ]
^G Ajuda      ^O Gravar     ^F Procurar   ^K Cortar     ^T Executar   ^C Posição
^X Sair       ^R Carregar   ^\ Substituir  ^U Colar      ^J Justificar ^/ Ir p/ linha
```

Output

```
kleiton-da-rosa@kleiton-da-rosa-Virtual-Machine:~/Secretária$ sudo bash /home/kl
eiton-da-rosa/Secretária/check_nginx.sh
quarta 10 jun 2026 20:50:14 -01 - Nginx está a funcionar.
```

9. Conclusão

A monitorização contínua, associada a uma política simples de backup e recuperação, permite reduzir o risco de indisponibilidade do serviço web. A utilização de verificações periódicas, consulta de logs e cópias de segurança garante que o serviço possa ser recuperado rapidamente em caso de falha, contribuindo para a continuidade operacional e para uma gestão mais segura da infraestrutura.